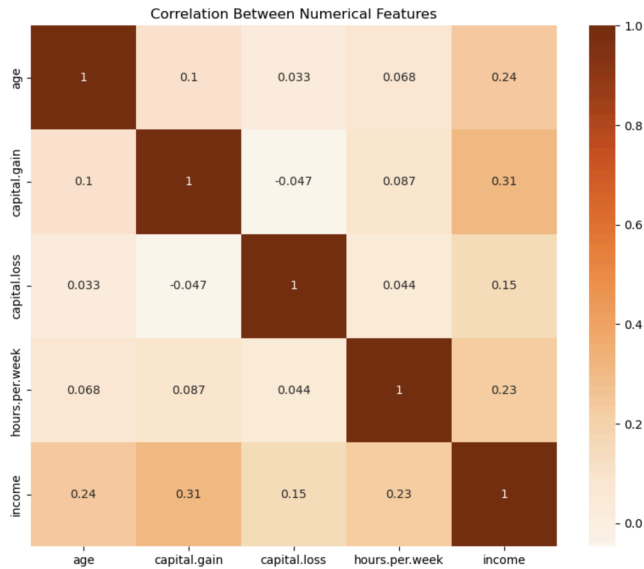


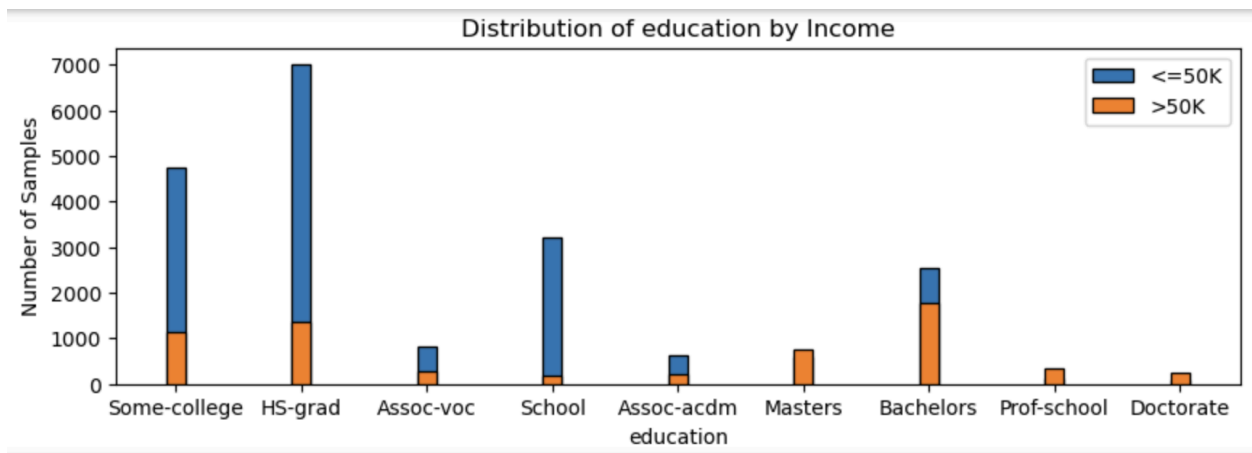
proj1_report

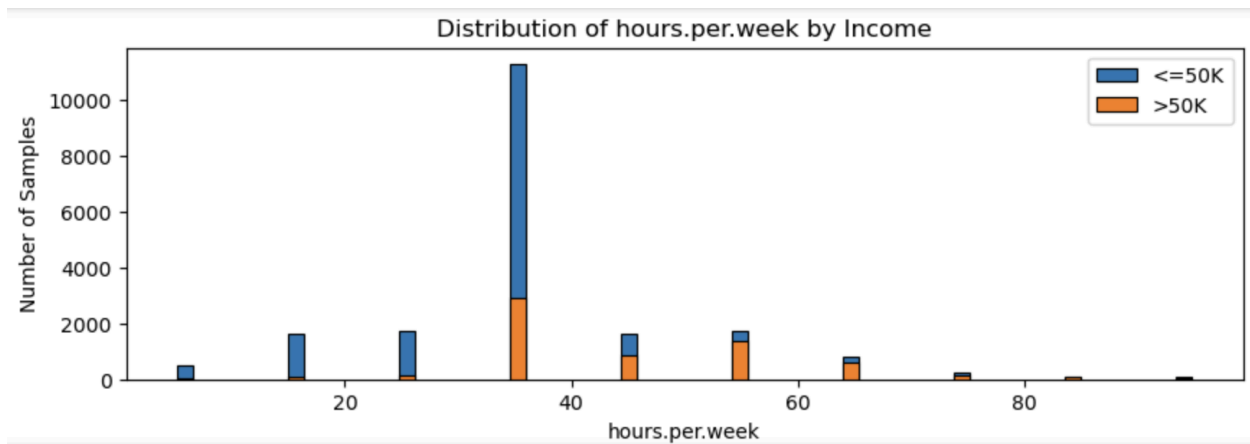
1.



From this heatmap, it appears that the strongest positive correlation is between income and capital gain (0.31), which would make sense as those with a larger income would typically see a larger capital gain. The weakest correlation in regards to income would be capital loss (0.15), which also checks out with the idea that those with higher incomes would see lower capital loss. Along with this, age appears to have a positive correlation with income (0.24), suggesting that older people tend to have a higher income.

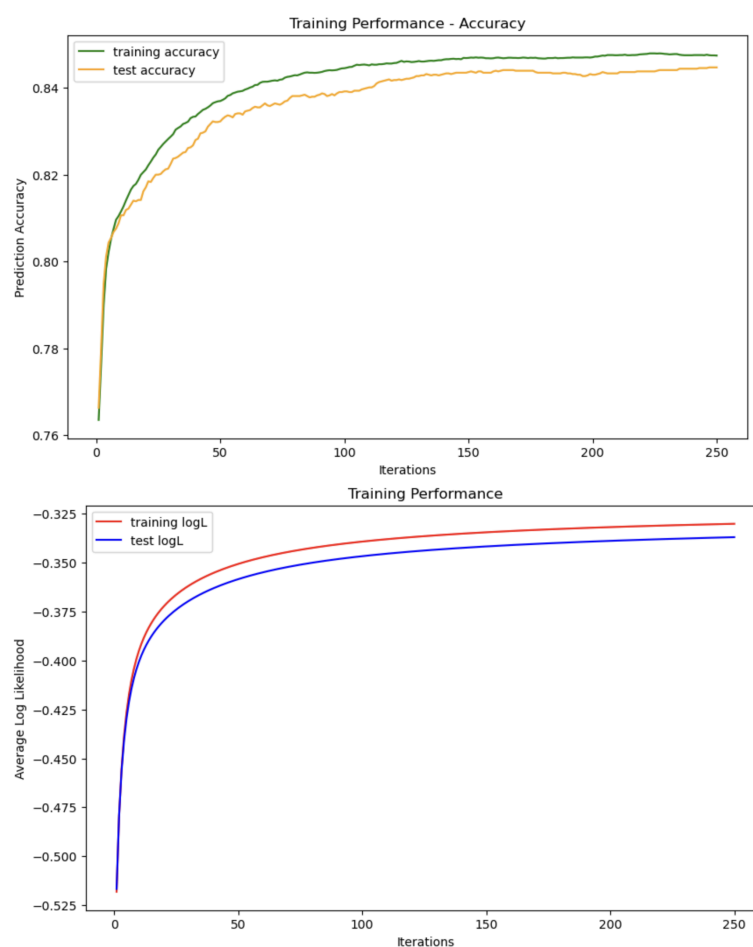
2.



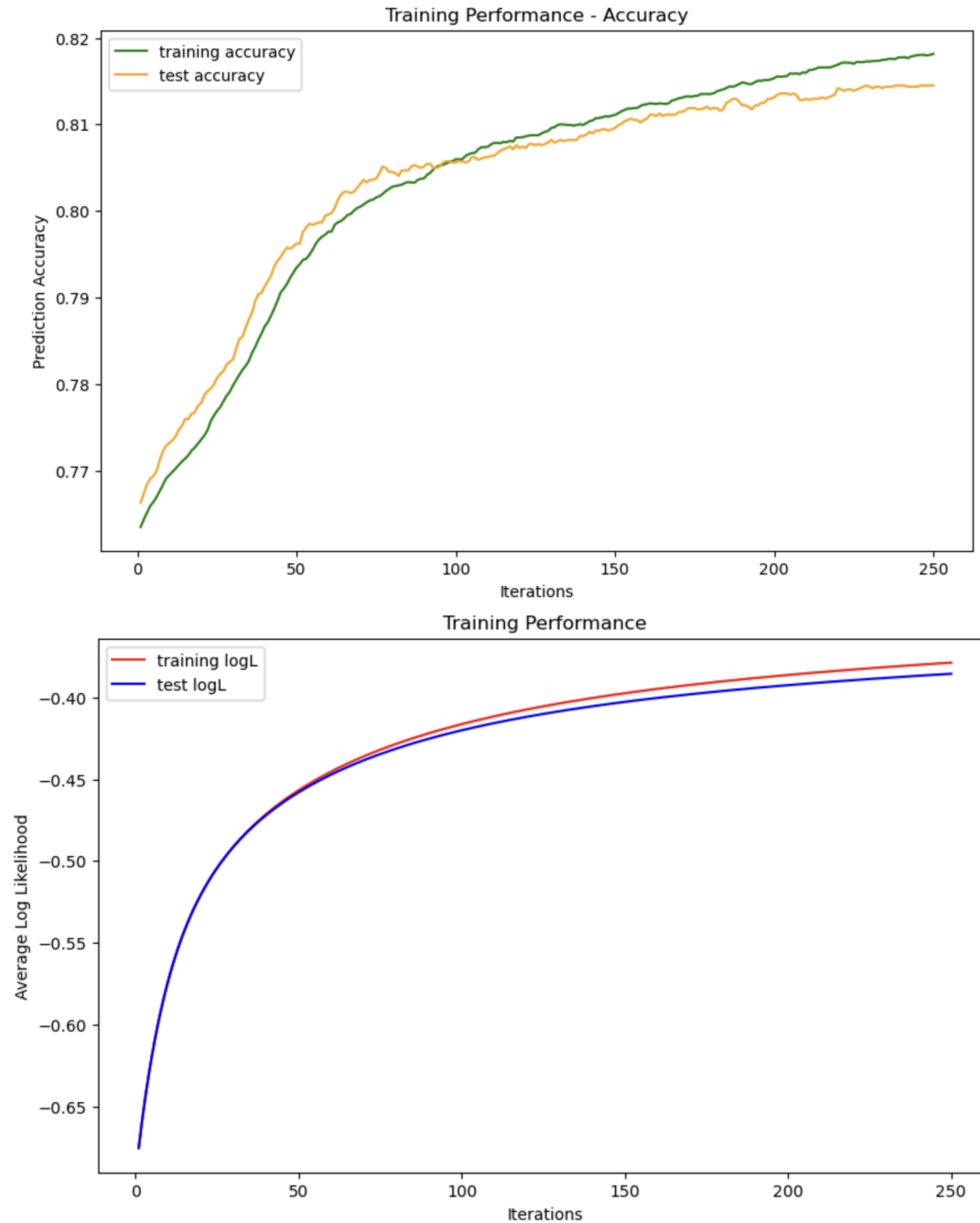


By analyzing the distribution of education by income, it appears that there were considerably more samples that made less than or equal to 50K across all levels of education except masters, prof-school, and doctorate. This suggests that those with a masters, prof-school, or doctorate tend to make more than 50K a year compared to those with less schooling. When analyzing the distribution of hours per week by income, it appears that those who make more than 50K tend to work at least about 30 hours a week, and overall, most samples work just under around 40 hours a week. This is to be expected as in most western countries, the average workweek is around 40 hours per week.

3. Learn Rate = 0.75

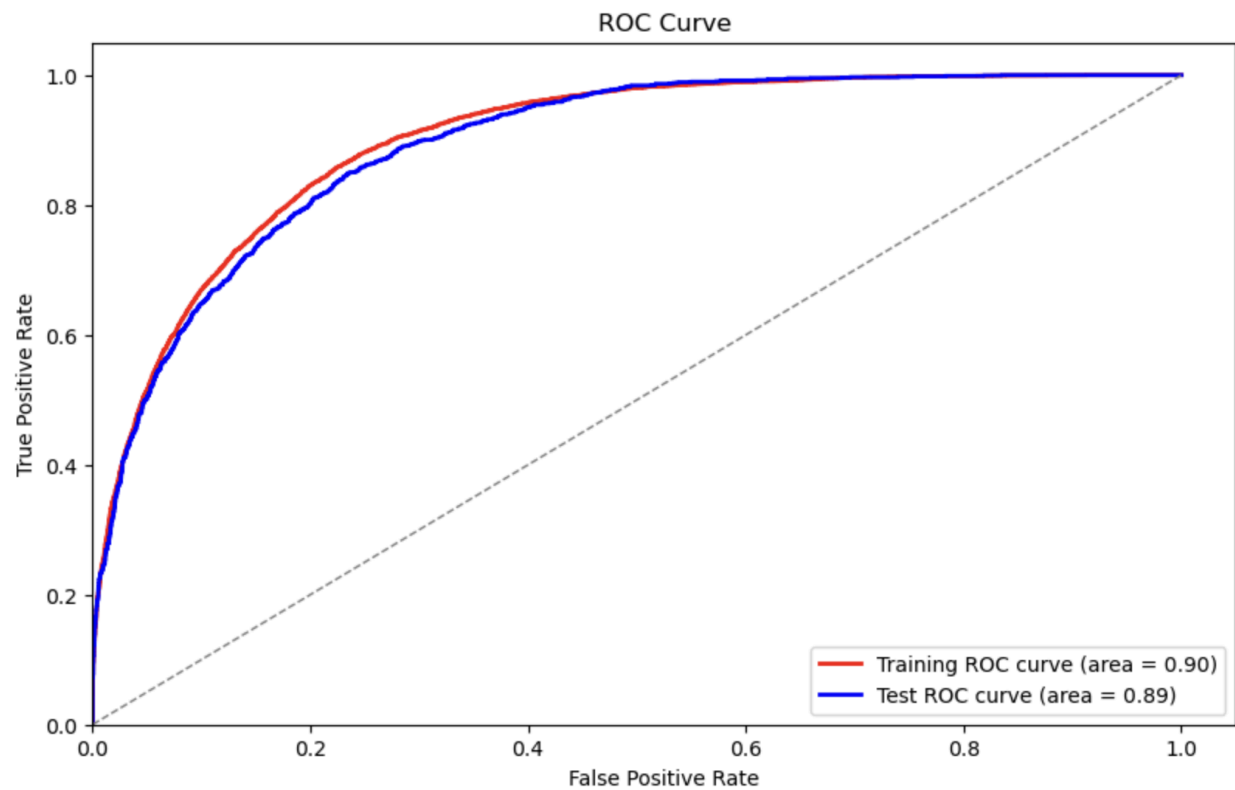


Learn Rate = 0.05



When analyzing the figures for prediction accuracy at a learning rate of 0.75 compared to 0.05, it appears that a learning rate of 0.75 reaches a greater prediction accuracy in many fewer iterations (0.84 vs 0.80 at around 100 iterations). This same trend can be seen when looking at the figures for the average log likelihood as it appears to increase at a much greater rate compared to that with a learning rate of 0.05. From this, I can conclude that a learning rate of 0.75 is superior to that of 0.05.

4. Learning Rate = 0.75



Learning Rate = 0.05

